
Signal Diode Reverse Characteristics

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% Soleil Demick
% ENGS 60 Lab 1

close all; clear; clc

% Read data
data = readtable("data/signal_reverse.csv");
Vr = data.Var2(3:end);
Ir = data.Var3(3:end)*1e-7*1e9; % 100nA/V -> nA

% Clean negative and nan values from data
nanmask = ~(isnan(Vr) | isnan(Ir)); % remove all rows with nan values
posmask = Vr > 0 & Ir > 0; % remove negative values
Vr = Vr(nanmask & posmask);
Ir = Ir(nanmask & posmask);

% Assume  $I = a \cdot V^b$ , solve for a,b
[powfit, gof] = fit(Vr, Ir, @(a, b, x) a.*x.^b)
powmodel = powfit(sort(Vr));

% % Linear plot
fig1 = figure;
hold on; grid on
% data
plot(Vr, Ir, LineWidth=2)
% power model
plot(sort(Vr), powmodel, LineWidth=2, Color="#6B6666", LineStyle="--")

% Labels
text(1, 3, sprintf("I_r = %fV_r^{%f}", powfit.a, powfit.b), FontSize=16)
ax = gca;
ax.FontSize = 16;
legend("Reverse IV Characteristic", "Power Model", ...
    Location="southeast")
xlabel("\bfReverse Voltage (V)")
ylabel("\bfReverse Leakage Current (nA)")
title("\bfSignal Diode Reverse Characteristics")
saveas(fig1, "graphics/signal_reverse_linear.png")

% % Logy plot
fig2 = figure;
% data
semilogy(Vr, Ir, LineWidth=2)
hold on; grid on
% power model
plot(sort(Vr), powmodel, LineWidth=2, Color="#6B6666", LineStyle="--")

% Labels
text(1, 3, sprintf("I_r = %fV_r^{%f}", powfit.a, powfit.b), FontSize=16)
```

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ax = gca;
ax.FontSize = 16;
legend("Reverse IV Characteristic", "Power Model", ...
      Location="southeast")
xlabel("\bfReverse Voltage (V)")
ylabel("\bfReverse Leakage Current (nA)")
title("\bfSignal Diode Reverse Characteristics")
saveas(fig2, "graphics/signal_reverse_log.png")

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Warning: Start point not provided, choosing random start point.

powfit =

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General model:
powfit(x) = a.*x.^b
Coefficients (with 95% confidence bounds):
a =      3.409   (3.405, 3.413)
b =      0.1447  (0.1433, 0.146)

```

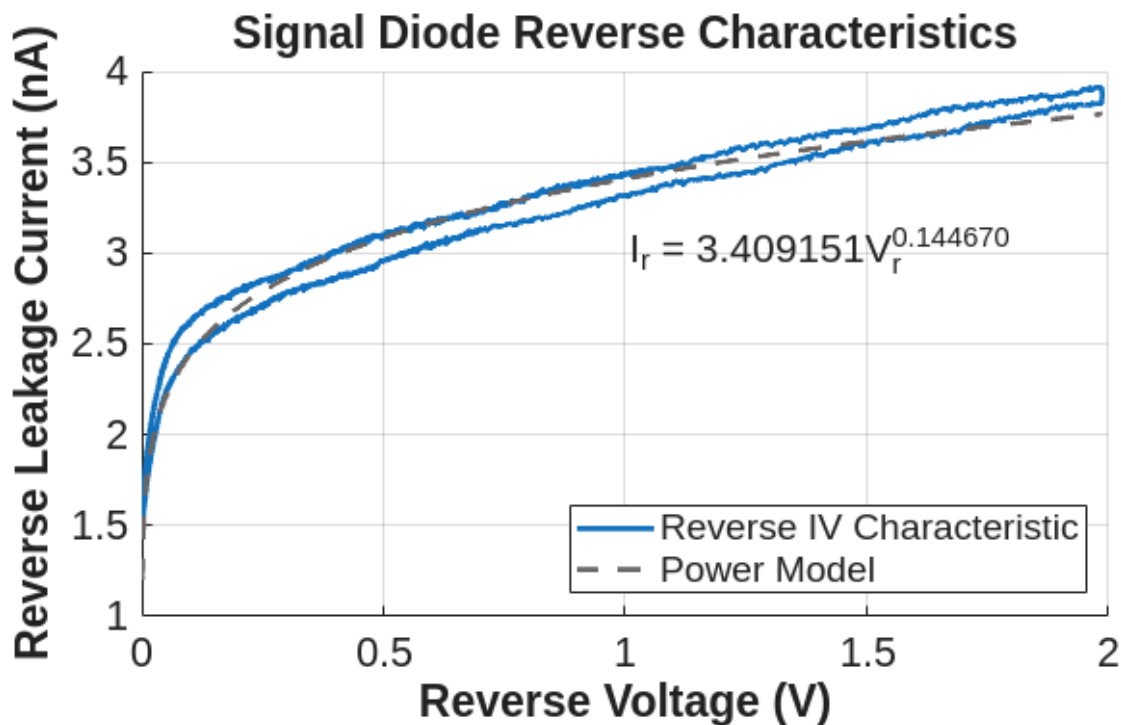
gof =

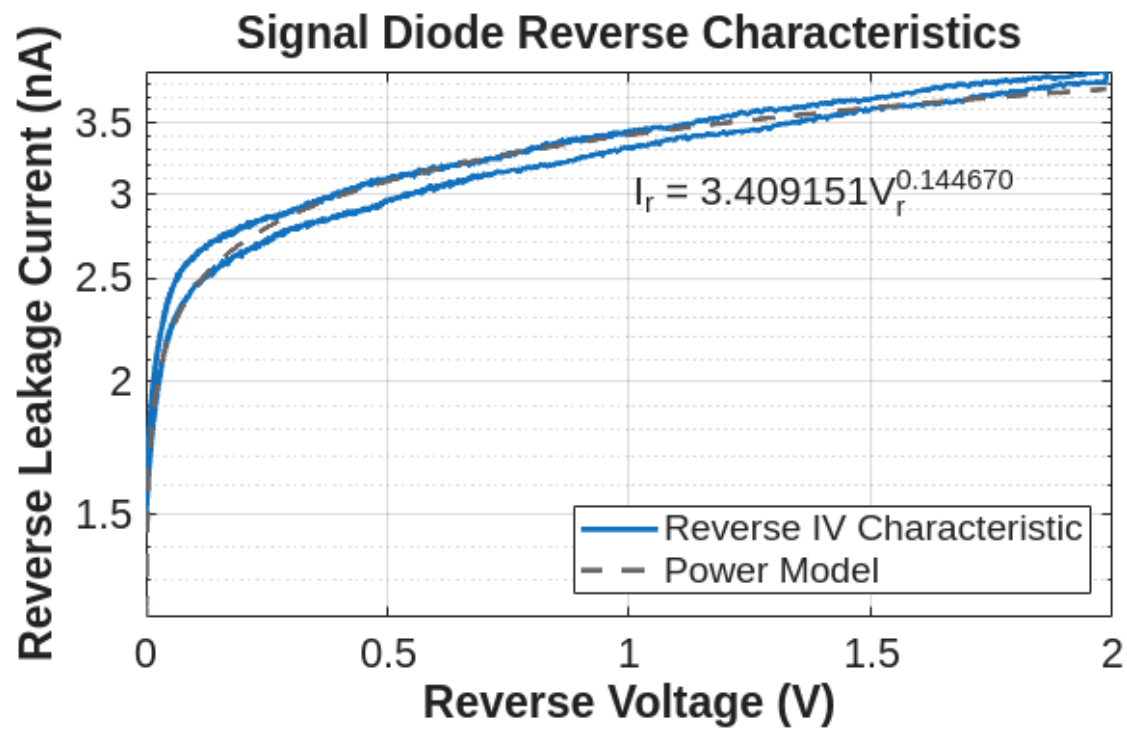
struct with fields:

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sse: 14.0125
rsquare: 0.9636
dfe: 1961
adjrsquare: 0.9636
rmse: 0.0845

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